

Compound Measures

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Complete the formula.

$$\text{speed} = \square \div \square$$

.....

2 [2 marks]

A car travels 180 km in 3 hours.

Work out the average speed.

.....

3 [2 marks]

A cyclist rides at an average speed of 15 km/h for 2 hours.

How far does the cyclist travel?

.....

4 [2 marks]

A train travels 240 km at an average speed of 80 km/h.

How long does the journey take?

.....

5 [3 marks]

A runner covers 400 m in 50 seconds.

(a) Work out the speed in m/s.

(b) Convert your answer to km/h.

.....

6

[3 marks]

A block of wood has mass 480 g and volume 60 cm^3 .
Work out its density.

7

[3 marks]

Gold has a density of 19.3 g/cm^3 .
Work out the mass of a piece of gold with volume 5 cm^3 .

8

[3 marks]

A journey of 90 km is driven at an average speed of 40 km/h.
Work out the time taken, in hours and minutes.

9

[4 marks]

A driver travels 60 km at an average speed of 30 km/h, then 90 km at an average speed of 45 km/h.
(a) Work out the total time for the journey.
(b) Work out the average speed for the whole journey.

10

[4 marks]

A box has weight 240 N. Its base measures 0.4 m by 0.3 m.
(a) Work out the area of the base.
(b) Work out the pressure the box exerts on the floor.
(c) The box is turned so that a face measuring 0.4 m by 0.2 m is on the floor. Work out the new pressure.

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1 [2 marks]

Convert 20 m/s into km/h.

.....

2 [2 marks]

Convert 90 km/h into m/s.

.....

3 [3 marks]

A car travels 45 miles in 50 minutes.
Work out the average speed in miles per hour.

.....

4 [3 marks]

A metal cube of side 4 cm has mass 1.2 kg.
Work out the density of the metal in g/cm^3 .

.....

5 [4 marks]

200 cm^3 of liquid A with density 0.8 g/cm^3 is mixed with 300 cm^3 of liquid B with density 1.2 g/cm^3 .
Work out the density of the mixture.

.....

6 [4 marks]

A train travels at a constant speed of 144 km/h.
How far does it travel in 25 seconds?

.....

7

[4 marks]

A country has an area of $2.4 \times 10^5 \text{ km}^2$ and a population of 1.8×10^7 .

- (a) Work out the population density in people per km^2 .
 - (b) The population increases by 20%. Work out the new population density.
-

8

[4 marks]

A tap fills a 45 litre bath in 6 minutes.

- (a) Work out the rate of flow in litres per minute.
 - (b) Work out the rate of flow in litres per second.
 - (c) How long would the tap take to fill a 300 litre tank?
-

9

[4 marks]

A cyclist rides 24 km at an average speed of 16 km/h, then returns along the same route at an average speed of 12 km/h.

Work out the average speed for the whole journey, to 2 decimal places.

10

[5 marks]

A solid metal cylinder has radius 3 cm and height 10 cm.

The metal has density 7.8 g/cm^3 and costs £0.45 per gram.

- (a) Work out the volume of the cylinder, to 3 decimal places.
 - (b) Work out the mass of the cylinder, to the nearest gram.
 - (c) Work out the cost of the metal, to the nearest penny.
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